

Tandem catalytic enantio- and diastereoselective synthesis of cyclopropyl alcohols using aryl aldehydes

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The prevalence of cyclopropanes in biologically active compounds has fueled many investigations into their preparation. Despite significant advances, more efficient methods for their catalytic enantioselective synthesis remain in demand. Previously, we reported a novel tandem approach to diastereo- and enantioenriched cyclopropyl alcohols. Our method involved an initial asymmetric C–C bond formation by addition of organozinc reagents to aldehydes catalyzed by a (–)-MIB-based catalyst. The resulting allylic zinc alkoxides were then subject to directed cyclopropanations. Although this approach provided cyclopropyl alcohols with very high enantio- and diastereoselectivity, it was successful with only aliphatic aldehydes. Aryl aldehyde substrates, on the other hand, exhibited low conversion and variable diastereomeric ratios. The present study is aimed at raising the stereoselectivity and yield of the aforementioned tandem reaction with aryl aldehyde substrates to make the method synthetically useful. Herein, we report the successful optimization of aryl aldehyde substrates in our one-pot tandem approach leading to cyclopropyl alcohols with high yields and enantio- and diastereoselectivities. Copyright © 2012 John Wiley & Sons, Ltd.

Supporting information may be found in the online version of this paper.

Keywords: cyclopropanation; cyclopropyl alcohols

INTRODUCTION

Cyclopropanes, the smallest strained carbocycles, have long been a target for organic chemists due to their abundance in biologically active compounds^[1–3] and their usefulness as intermediates in organic transformations.^[4,5] Not surprisingly, their unusual structural/chemical properties and interesting bonding characteristics have attracted the attention of physical organic chemists.^[6] Despite their strained structures and high reactivity, cyclopropanes are usually quite stable at room temperature, and are present in a variety of natural and unnatural products including pheromones, terpenes, fatty acid metabolites and epothilone analogs.^[1–3,7–10]

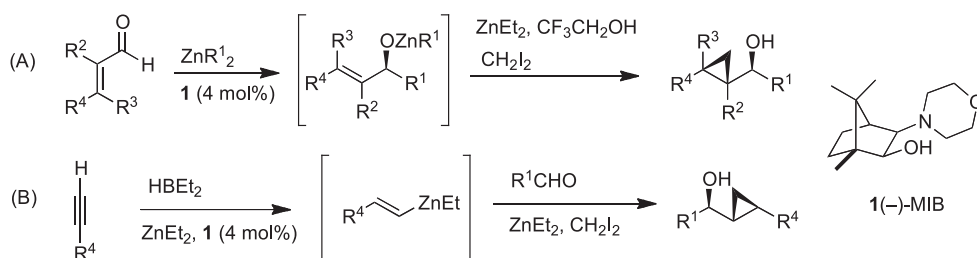
In recent years, a growing interest in enantioenriched cyclopropanes has stirred the development of their asymmetric synthesis.^[11] The Simmons-Smith cyclopropanation has been frequently employed in the stereoselective cyclopropanation of olefins and allylic alcohols with chiral auxiliaries or with stoichiometric additives, and excellent stereoselectivities have been reported.^[12,13] However, the development of catalytic enantioselective Simmons-Smith cyclopropanations remains challenging.^[14–26] In particular, the asymmetric cyclopropanation of allylic alcohols suffers from suboptimal enantioselectivities and high catalyst loadings with moderate substrate scope.^[14–21,27] Diastereoselective cyclopropanation of enantioenriched allylic alcohols also gives rise to enantio- and diastereoenriched cyclopropyl alcohols, but requires the synthesis and purification of optically active secondary alcohols.^[20,28]

Recently, we reported a tandem catalytic enantio- and diastereoselective synthesis of cyclopropyl alcohols starting from *achiral* reagents, thus eliminating the need for isolation of enantioenriched allylic alcohols.^[29–31] Our general strategy involved catalytic

enantioselective carbonyl additions to aldehydes followed by diastereoselective cyclopropanation of the resulting allylic zinc alkoxide intermediates (Scheme 1). Performing these reactions in tandem enabled the formation of three C–C bonds and three stereocenters in a one-pot procedure with excellent stereoselectivities. In order to broaden the substrate scope, the allylic zinc alkoxides were generated via two routes; an asymmetric alkyl addition to α,β -unsaturated aldehydes in the presence of 4 mol% of Nugent's (–)-MIB ligand^[32,33] (Scheme 1, route A), or an asymmetric addition of a vinyl group, generated using the Oppolzer/Srebnik method,^[34–36] to saturated aldehydes (Scheme 1, route B). Through these two approaches, a variety of cyclic and acyclic *cis*- and *trans*-disubstituted cyclopropyl alcohols could be accessed with high stereoselectivities. Route B, however, exhibited some significant substrate limitations when aryl aldehydes were employed. These substrates were characterized by low conversions and variable diastereoselectivities despite many attempts to optimize the process. Herein, we present our efforts to overcome this significant substrate limitation and report a highly diastereoselective one-pot cyclopropanation of allylic zinc alkoxide intermediates derived from asymmetric vinylation of *aromatic* aldehydes.

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Scheme 1. Two tandem asymmetric addition/diastereoselective cyclopropanations

Experimental section

Experimental details and full characterization of all compounds are provided in the Supporting Information.

General procedure for the synthesis of cyclopropyl alcohols

Alkynes (0.65 mmol) and diethylborane (0.65 mmol, 1.0 M in toluene) were added to a dry Schlenk flask under nitrogen and stirred at room temperature for 30 min. After the reaction flask was cooled to -78°C , (–)-MIB (0.02 mmol, 4 mol %) was added followed by Et_2Zn (0.75 mL, 1.0 M in hexanes). The reaction mixture was then warmed to -10°C , and aryl aldehydes (0.5 mmol) were added dropwise. The solution was stirred at this temperature until the vinyl addition was complete by TLC (~8 h). The volatile materials, including the byproduct Et_3B , were removed in vacuo at 0°C . Hexanes (2.0 mL) was added, and the volatile materials were again removed under reduced pressure. This step was repeated two more times to insure removal of Et_3B . A solution of Et_2Zn (2.5 mmol, 2.5 M in dichloromethane) and neat $\text{CF}_3\text{CH}_2\text{OH}$ (2.5 mmol) were added at 0°C . After stirring for 5 min at 0°C , diiodomethane (2.5 mmol) was added dropwise, and the reaction mixture was allowed to room temperature. The Schlenk flask was covered with aluminum foil to exclude light, and the reaction mixture was stirred for 24 h. The reaction was then quenched with saturated aq. NH_4Cl (15 mL). The organic and aqueous layers were separated, and the aqueous layer was extracted with 3X20 mL dichloromethane. The combined organic layers were then washed with saturated aqueous NaHCO_3 and brine and dried over MgSO_4 . The filtrate was concentrated in vacuo, and the residue was chromatographed on silica to afford the title compound.

The following compounds were made by the General Procedure outlined above:

(2-Butylcyclopropyl)(phenyl)methanol (**1a**): ^1H NMR (CDCl_3 , 300 MHz): δ 7.37 (m, 5 H), 4.06 (d, 1 H, $J=8.43$ Hz), 1.92 (br s, 1 H), 1.20 (m, 5 H), 0.94 (m, 2 H), 0.81 (t, 3 H, $J=6.64$ Hz), 0.76 (m, 1 H), 0.65 (m, 1 H), 0.44 (m, 1 H); ^{13}C NMR (CDCl_3 , 75 MHz): δ 140.0, 128.6, 127.8, 126.2, 78.5, 33.5, 31.8, 27.3, 22.6, 17.3, 14.3, 11.5.

(2-*tert*-Butylcyclopropyl)(*p*-tolyl)methanol (**1b**): ^1H NMR (CDCl_3 , 500 MHz): δ 7.21 (d, 2 H, $J=7.85$ Hz), 7.08 (d, 2 H, $J=7.85$ Hz), 3.95 (d, 1 H, $J=8.07$ Hz), 2.28 (s, 3 H), 1.80 (br s, 1 H),

1.02 (m, 1 H), 0.64 (s, 9 H), 0.60 (m, 1 H), 0.49 (m, 1 H), 0.42 (m, 1 H); ^{13}C NMR (CDCl_3 , 125 MHz): δ 137.4, 129.1, 126.0, 78.7, 29.4, 28.7, 28.5, 23.4, 21.3, 7.0.

(2-*tert*-Butylcyclopropyl)(4-fluorophenyl)methanol (**1c**): ^1H NMR (CDCl_3 , 300 MHz): δ 7.37 (m, 2 H), 7.02 (m, 2 H), 3.98 (d, 1 H, $J=8.61$ Hz), 2.04 (br s, 1 H), 1.02 (m, 1 H), 0.71 (m, 1 H), 0.68 (s, 9 H), 0.57 (m, 1 H), 0.49 (m, 1 H); ^{13}C NMR (CDCl_3 , 75 MHz): δ 140.3, 127.9, 127.8, 115.5, 115.3, 78.7, 29.5, 29.0, 28.6, 24.0, 7.5.

(2-Butylcyclopropyl)(4-(trifluoromethyl)phenyl)methanol (**1d**): ^1H NMR (CDCl_3 , 300 MHz): δ 7.56 (m, 4 H), 4.10 (d, 1 H, $J=8.56$ Hz), 2.01 (br s, 1 H), 2.10 (m, 4 H), 0.92 (m, 3 H), 0.80 (t, 3 H, $J=6.80$ Hz), 0.78 (m, 1 H), 0.67 (m, 1 H), 0.45 (m, 1 H); ^{13}C NMR (CDCl_3 , 75 MHz): δ 145.9, 128.0, 127.8, 126.5, 125.6, 77.9, 33.3, 31.8, 27.6, 22.6, 17.6, 14.3, 11.1.

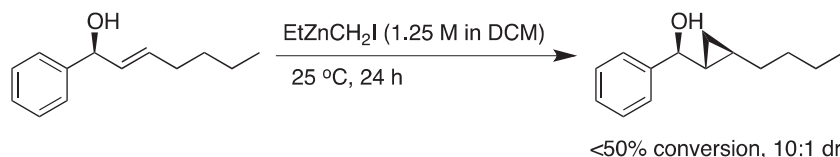
(2-*tert*-Butyl-1-methylcyclopropyl)(phenyl)methanol (**1e**): ^1H NMR (CDCl_3 , 300 MHz): δ 7.37 (m, 5 H), 3.99 (s, 1 H), 1.85 (br s, 1 H), 1.08 (s, 3 H), 0.93 (s, 9 H), 0.89 (m, 1 H), 0.66 (dd, 1 H, $J=4.45$, 9.84 Hz), 0.43 (dd, 1 H, $J=4.45$, 6.68 Hz); ^{13}C NMR (CDCl_3 , 75 MHz): δ 142.9, 128.1, 127.4, 126.5, 83.0, 34.4, 30.8, 30.6, 28.0, 14.3, 13.4.

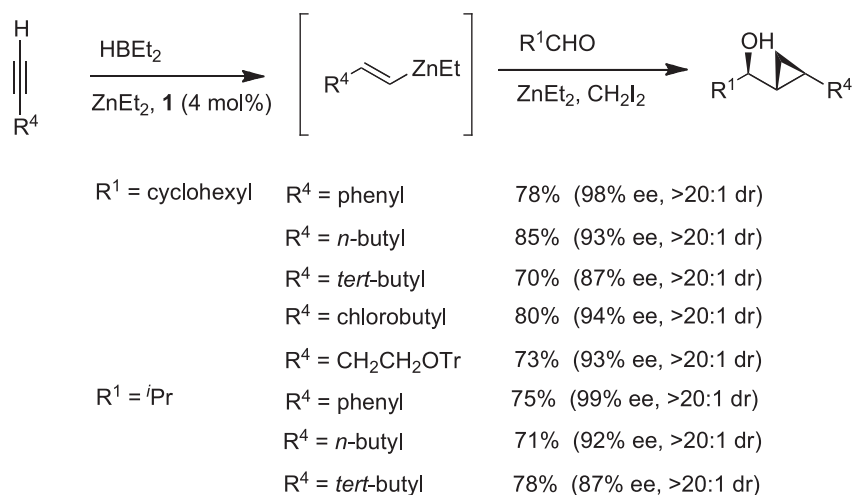
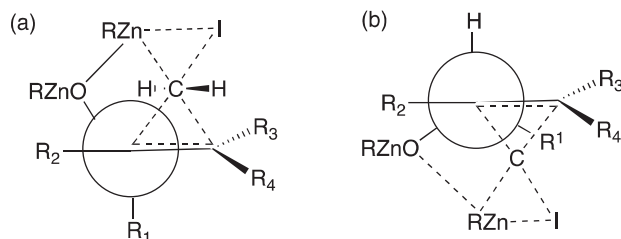
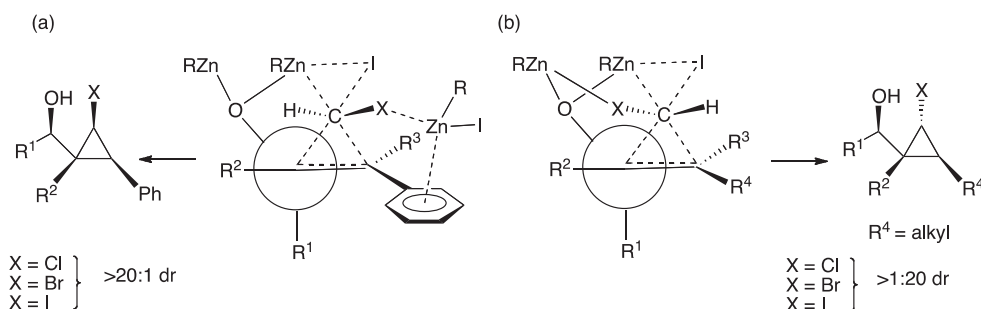
(2-*tert*-Butylcyclopropyl)(2-bromophenyl)methanol (**1f**): ^1H NMR (CDCl_3 , 500 MHz): δ 7.53 (m, 2 H), 7.34 (m, 1 H), 7.12 (m, 1 H), 4.62 (d, 1 H, $J=7.73$ Hz), 2.07 (br s, 1 H), 1.12 (m, 1 H), 0.93 (m, 1 H), 0.70 (s, 9 H), 0.50 (m, 2 H); ^{13}C NMR (CDCl_3 , 125 MHz): δ 143.6, 133.0, 129.1, 128.4, 122.6, 78.2, 29.5, 28.5, 27.9, 22.6, 7.0.

(2-*tert*-Butylcyclopropyl)(*o*-tolyl)methanol (**1g**): ^1H NMR (CDCl_3 , 500 MHz): δ 7.44 (d, 1 H, $J=7.50$ Hz), 7.15 (t, 1 H, $J=7.50$ Hz), 7.10 (t, 1 H, $J=7.50$ Hz), 7.06 (d, 1 H, $J=7.50$ Hz), 4.40 (d, 1 H, $J=7.70$ Hz), 2.31 (s, 3 H), 1.69 (br s, 1 H), 0.81 (m, 1 H), 0.66 (s, 9 H), 0.62 (m, 1 H), 0.49 (m, 1 H), 0.35 (m, 1 H); ^{13}C NMR (CDCl_3 , 125 MHz): δ 142.4, 142.0, 130.5, 127.4, 126.3, 126.2, 74.3, 29.6, 28.5, 28.0, 21.9, 19.6, 6.7.

RESULTS AND DISCUSSION

To delineate the cause of the low reactivity and variable selectivity observed in our one-pot vinylation/cyclopropanation of benzaldehyde derivatives (Scheme 1B), we examined the cyclopropanation of an isolated allylic alcohol. Subjecting the allylic alcohol to our cyclopropanation conditions, which were successfully used with aliphatic aldehydes, resulted in generation of cyclopropyl alcohols with low conversion (< 50%) and with 10:1 dr, as shown in Eqn 1.



**Scheme 2.** Asymmetric vinyl addition/diastereoselective cyclopropanation**Figure 1.** Transition states of cyclopropanation: (a) *syn*-selective directed/favored transition state and (b) *anti*-selective directed/disfavored transition state**Figure 2.** Zinc- π interaction in the transition state of halocyclopropanation

This result led us to hypothesize that the low yields in the tandem addition/cyclopropanation were due to the cyclopropanation step and not associated with the formation of the allylic zinc alkoxide in Scheme 1B. We had previously encountered a similar problem, but with a different origin, in our tandem vinylation of aliphatic aldehydes/cyclopropanation (Scheme 1B). In this case, the poor reactivity was caused by the triethylborane byproduct, formed during the transmetalation step, which interfered with the cyclopropanating reagents.^[29] An experimental modification involving

the removal of volatile materials, including the triethylborane byproduct, after transmetalation provided the cyclopropyl alcohols with excellent stereoselectivities (Scheme 2).^[29] This modified cyclopropanation procedure was successful with aliphatic aldehydes that failed to improve the reaction outcomes of *aromatic* aldehyde substrates.

In the case of aromatic aldehyde substrates, it is possible that interactions between the cyclopropanating reagents and benzylic alkoxide result in the decreased reactivity. The widely accepted 'butterfly-type' transition state of the Simmons-Smith

cyclopropanation, providing the *syn*-diastereomer as a major product, is illustrated in Fig. 1a.^[13] The *syn*-pathway is strongly favored over the *anti*-pathway (Fig. 1b) due to unfavorable interactions between R¹ and R³. Although we do not know the nature of the interaction of the zinc carbenoid with the aryl π -system, it is noteworthy that several reports describe favorable interactions between Zn(II) and unsaturated π -systems.^[30,37–42] For example, we had previously shown that in the halocyclopropanation reaction, the zinc carbenoid interacts with aryl groups at R⁴, resulting in a reversal in diastereoselectivity (Fig. 2a).^[30] In contrast, when R⁴ = alkyl, the opposite diastereomer was obtained with excellent diastereoselectivity (Fig. 2b). Decreasing the electron density in the aryl ring resulted in a decrease in the diastereomeric ratio and is consistent with a Zn- π -interaction. It is also known that Zn(C₆F₅)₂ forms a π -complex with toluene, as determined by X-ray crystallography.^[41]

Given that the interaction of Zn(II) with unsaturated π -system of benzylic alkoxide intermediate could be the root of problem, we envisioned increasing the reactivity of the cyclopropanating reagents. We were particularly encouraged by Charette's findings illustrating the importance of the nature and stoichiometry of cyclopropanating reagents in the diastereofacial control in the cyclopropanation of chiral allylic alcohols.^[43] Thus, we revisited our tandem asymmetric vinylation of *aromatic* aldehydes followed by diastereoselective cyclopropanation. First, the use of zinc carbenoid species with different electronic properties was studied (Table 1). It is known that electron withdrawing additives, such as CF₃CO₂H,^[44,45] CF₃CH₂OH,^[29–31,44,45] and phenol derivatives,^[46] can generate more reactive zinc carbenoids of the type Y–ZnCH₂I (Y is the conjugate base of the acids listed above). Marginal improvement in the cyclopropanation conversion was observed, in particular with use of CF₃CH₂OH (entry 2). Simply raising the temperature to 40 °C did not result in any beneficial effect, possibly due to the increased rate of carbenoid decomposition (entry 4). Next, the concentration of carbenoid was examined focusing on the additive CF₃CH₂OH. By increasing the concentration of the carbenoid, we hoped to raise the rate of the cyclopropanation

over the rate of carbenoid decomposition. Gratifyingly, reactions conducted under more concentrated conditions (i.e. 2.5 M of zinc carbenoids as opposed to 1.25 M) showed a marked improvement in the reaction rate with excellent yield and diastereoselectivity (entry 5). Further increasing the concentration resulted in decreased yield and formation of significant byproducts (entry 6).

Our optimized conditions were then applied to other allylic zinc alkoxide intermediates generated by vinylation of a variety of *aromatic* aldehyde substrates. Electron-rich or -poor *aromatic* aldehydes with *para* substituents gave products with high ee's and excellent dr's (Table 2, entry 1–4). Our method was also successful in the preparation of trisubstituted cyclopropyl alcohols with high stereoselectivities and good yield. Thus, regioselective hydroboration of the internal alkyne 4,4-dimethylpent-2-yne followed by asymmetric addition and cyclopropanation furnished the product in 82% yield with 93% ee and 20:1 dr (entry 5). Surprisingly, low diastereoselectivity was observed with 2-bromobenzaldehyde (~1.7:1 dr, entry 6). We reasoned that a halogen atom interferes with the alkoxide directed transition state in the cyclopropanation by interaction of the zinc carbenoid with the bromide substituent. It is known that allylic oxygen-containing groups exert a directing effect in the cyclopropanation.^[47] For example, the directing ability of ethers, esters, and acetals in the Simmons-Smith has been documented.^[48–51] The directing effect of halogen atoms in the cyclopropanation has not been previously observed to our knowledge. Indirect evidence for the interaction of the bromide substituent was gained upon subjecting 2-methylbenzaldehyde to our reaction conditions, where a single diastereomeric cyclopropyl alcohol was isolated (entry 7).

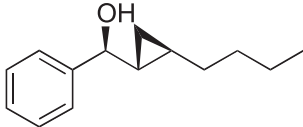
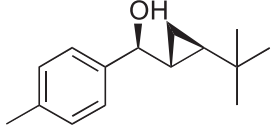
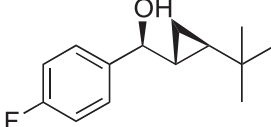
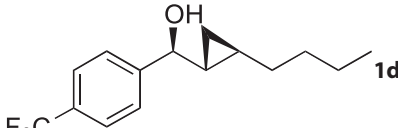
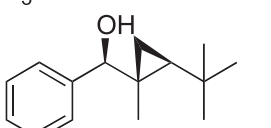
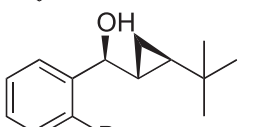
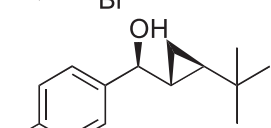
In conclusion, we have successfully extended the substrate scope of our one-pot tandem vinyl addition/diastereoselective cyclopropanation method to *aromatic* aldehydes. One key challenge herein was to accelerate the cyclopropanation reaction rate while minimizing the potential Zn(II)- π interactions. Upon screening several reaction parameters, i.e. additives,

Table 1. Optimization of cyclopropanation step

entry	Additive	temperature (°C)	conversion (%) ^a	comment
1 ^b	CF ₃ CO ₂ H	25	50	
2 ^b	CF ₃ CH ₂ OH	25	60	
3 ^b	2,4,6-trichlorophenol	25	35	
4 ^b	CF ₃ CH ₂ OH	40	<20	not complete and many products
5	CF ₃ CH ₂ OH	25	80 ^c	2.5 M, >20:1 dr
6	CF ₃ CH ₂ OH	25	50 ^c	4.0 M, reaction complete but many products

^aDetermined by ¹H NMR analysis of the crude reaction mixture.
^bConcentration of zinc carbenoids: 1.25 M, inconsistent dr (10:1 ~ >20:1) was observed
^cIsolated yield

Table 2. Tandem asymmetric vinyl addition to *aromatic* aldehydes/diastereoselective cyclopropanation

Entry	cyclopropyl alcohols	yield (%) ^a	ee (%) ^b	dr ^c
1	 1a	80	94	>20:1
2	 1b	80	95	>20:1
3	 1c	73	93	>20:1
4	 1d	71	95	>20:1
5	 1e	82	93	>20:1
6	 1f	72 ^d	93	~1.7:1
7	 1g	82	92	>20:1

^aIsolated yield.
^bDetermined by HPLC.
^cDetermined by ¹H NMR analysis of the crude reaction mixture.
^dCombined yield of two diastereomers.

temperature, and concentration, the reaction concentration was found to be crucial with Shi's cyclopropanating reagent ($\text{CF}_3\text{CH}_2\text{OZnCH}_2\text{I}$). A variety of *aromatic* aldehyde substrates were successfully employed under our optimized conditions. Given the efficiency and high diastereoselectivity of our cyclopropanation approach, we anticipate this tandem addition/cyclopropanation will serve as a useful synthetic tool in organic synthesis.

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